BOM STANDARDISATION

PROJECT BRIEF

PROJECT GROUP: CONTROL SYSTEMS

POC

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Project Overview

The current car has 14 custom printed circuit boards (PCBs). Each PCB has a Bill-of-materials (BOM) that specifies what components are used. This project proposes the creation of a single document to keep track of the BOMs of all the boards. This document will serve to inform the creation of a control systems department-level component selection standard.

Task Description

The BOM Standardisation project involves the creation of an Excel document to show the 14 boards in the car and the list of all components used across them and should include the following:

- Quantity of each component on each board
- Total number of components on each board
- Total number of unique components on each board
- Classify each component into a group (resistor, capacitor, LED, etc.)
- Filterable by component, board and component class
- Support the easy addition of new boards or changes to an existing boards BOM

Deliverables

The completion of the BOM Standardisation project expects the creation of a completed "BOM Standardisation.xlsx" document. A document outlining how to use the document, add new boards, or change an existing boards BOM would be beneficial.

Get Started

Resources

Resource	Description
Google Drive | BOM Standardisation project folder	Folder containing the "BOM Standardisation" document
Google Drive | Board Documentation	Contains the board documentation folders
Altium Drive	Contains the board BOMs
Discord | #control-systems-general	Discord channel to ask questions for this project.

Plan

Each board on the Altium Drive contains an option to download the BOM.
Compile all 14 BOMs into a single document. Automation using scripting is highly recommended.
Modify the document to provide the functionality outlined in the Task Description heading.
The list of boards on the car is:

- ASSI/Brake Light
- BMU
- BSPD
- BSPD Pedal Sensor
- Cell sideboards
- CMU
- CMU Breakout
- Dash LED
- Discharge Board

VCU Characterisation

- MegaVCU
- Shutdown
- TSAL Control
- TSAL LED
- TSMP